CHAPTER 1

Marine biodiversity: its past development, present status, and future threats

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1.1 Introduction

'We do not associate the idea of antiquity with the ocean, nor wonder how it looked a thousand years ago, as we do of the land, for it was equally wild and unfathomable always. The Indians have left no traces on its surface, but it is the same to the civilized man and the savage. The aspect of the shore only has changed.'

—Henry David Thoreau; Cape Cod (1855–1865), in The Writings of Henry David Thoreau, vol. 4, p. 188, Houghton Mifflin (1906)

Humans are terrestrial creatures, and those with a deep knowledge of the sea tend to be those associated either with its exploitation or exploration. Our lack of knowledge about life in the marine realm, what lives there, where it lives, and what it does, should not be underestimated. Seawater covers around 70% of the Earth, and the vast majority of landscape beneath it has never been seen by humans, let alone sampled in any way. Sampling the oceans is costly and having to work from ships or boats puts many constraints on what can be done, where, and how often. The vast majority of marine biodiversity records are from shallow waters (< 200 m), or from the seabed, and the deep pelagic ocean, which represents the largest volume on Earth in which life can exist (> 10^9 km^3) is recognized as particularly under-represented in global databases (Webb *et al.* 2010). The total area of seabed surveyed by human eyes, in submersibles, by diving, or remotely using cameras, represents an infinitesimal fraction of the total. To compound this problem, the majority of marine life doesn't even live on the seabed where cameras can see it, but lives either buried within it or floating above it in the water column. In addition, much of the metazoan diversity in the water column consists of extremely delicate forms which are destroyed by conventional sampling using nets. Consequently, our knowledge of marine biodiversity has been likened to what our knowledge of rainforest biodiversity might be like if our only way of assessing it was by means of a grappling hook deployed from a hot-air balloon. Recent discoveries have highlighted the depth of our ignorance. These include the discovery in 1977 of hydrothermal vent communities on the seabed, deriving their energy from chemoautotrophy rather than sunlight (Lonsdale 1977), the identification in 1986 of *Prochlorococcus* in the photic zone, a tiny photosynthetic cyanobacterium which is probably the most abundant organism on Earth and is estimated to account for 20% of the oxygen in the Earth's atmosphere (Chisholm *et al*. 1988), and in the deep pelagic ocean, the discovery of the very large, charismatic and widely distributed megamouth shark *Megachasma pelagios* in 1976 (Berra 1997). Finally, but perhaps most notably, are the findings generated over the past 10 years by the Census of Marine Life (<http://www.coml.org>). This census has brought together nearly 3000 scientists from over 80 nations to assess the diversity, distribution and abundance of marine life. Their results indicate that there are approximately 250 000 valid marine (non-microbial) species formally described in the scientific literature with as many as another 750 000 species still waiting to be described. They also estimate that there may be more than a billion types of microbes living in the oceans. These recent, concerted, international

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activities have started to reveal the staggering enormity of marine biodiversity.

1.2 What is biodiversity?

The term biodiversity is a contraction of 'biological diversity' and first appeared in print in 1988 (Wilson and Peters 1988), very quickly becoming widespread (Haila and Kouki 1994). The use of this term arose amid concern over the loss of the natural environment and, as a result, it has become part of the lexicon of science, the media, and government on a global scale (Gaston 1996). The ultimate confirmation of this was the signing, in 1992, of the UN Convention on Biological Diversity (<http://www.cbd.int>). According to the convention biodiversity means 'the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems.' Gaston (1996) provides a number of alternative definitions which are essentially variations on this theme, and an interesting discussion of what biodiversity is, or might be. Essentially, biodiversity is a multivariate concept with 'the variety of life' distributed across a range of hierarchical scales and cannot, therefore, be boiled down to a single measure. For example, even with a defined biological category such as macrofauna (multicellular animals larger than 0.5 mm), diversity is not only a function of the number of species in any given area (known as richness), but also the way in which individuals are distributed within those species (evenness). There are even different measures of richness depending on whether you are interested in taxonomic or functional diversity. This provides a dilemma in situations where there is a desire to measure biodiversity and use it as an end point in rigorous empirical studies (Solbrig 1991; Harper and Hawksworth 1994). In practice, how biodiversity is measured will depend on the specific nature and purpose of the study in question. To date, much of the biodiversity research effort exploring its relevance to human governance, well-being, and survival has tended to focus on terrestrial systems, with the possible exception of fisheries.

1.3 Comparing marine and terrestrial biodiversity

Marine ecosystems are fundamentally different from terrestrial ecosystems (Steele 1985; Strathmann 1990). Biomass per unit area in marine systems may be several orders of magnitude less than in terrestrial or freshwater systems, but it must be remembered that terrestrial systems are essentially two-dimensional with most living material concentrated into a layer 10–100 m thick. The seas and oceans, on the other hand, have an average depth in excess of 3500 m, and two-thirds of the Earth's surface lie below more than 200 m of seawater. Where terrestrial primary producers tend to be large and sessile, marine primary producers tend to be small and mobile, and therefore subject to fluid transport processes and spatial mixing. Photosynthesis is limited to the upper sunlit parts of the ocean, which over vast areas are very much less productive than the land because growing phytoplankton strip nutrients from the surface waters which are removed as they sink and die. Essentially, therefore, much marine productivity is self-limiting. Production and consumption may be decoupled in the sea with fixed carbon being consumed many times before it is completely remineralized and, although available organic matter becomes increasingly scarce with depth, it supports diverse pelagic and benthic ecosystems. At the other end of the marine food-web, consumers, grazers, and carnivores, which may be several times larger than their terrestrial counterparts, have a much greater range of life-history characteristics than terrestrial ones. Many have planktonic and benthic stages, with different associated environmental linkages. Marine systems may be relatively 'open' with gene-flow between distant marine habitats facilitated by larval dispersal. Many marine predators have extremely high reproductive output especially among older and larger members of their populations, producing millions of offspring. This may help them to survive over-exploitation, at least for a while, but also makes populations highly variable, difficult to predict, and vulnerable to threshold effects.

Unsurprisingly, we do not actually know how many species currently live in Earth's seas and

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oceans although current estimates for multicellular organisms are in the order of 250 000 species (May 1988, Census of Marine Life). If correct, this is far less than the numbers estimated to live on land, but marine diversity, at least animal diversity, is organized very differently to its terrestrial counterpart. To put it glibly, to a first approximation, all species of land animals are insects. In contrast, marine biodiversity is more evenly distributed between the major animal groups. There are 35 different phyla that occur in the sea, with most of them being benthic (Briggs 1994, May 1988) and only 11 occurring in the water column (Angel 1993). From the total 35 phyla, 14 are known to occur only in the sea. This compares favourably with freshwater systems, containing only 14 phyla, and terrestrial systems that can only boast representatives from 11 phyla. Thus animal phyletic diversity is clearly at its highest in the marine realm.

As a consequence of these fundamental differences, biodiversity paradigms developed in terrestrial systems may not be transferable to marine systems and the response of these two systems to human perturbations will be markedly different (National Research Council 1995).

1.4 The rise of marine biodiversity

Until recently, our knowledge of the origins and evolution of life on Earth was primarily derived from the fossil record, but currently our estimates of the timing and nature of major events are being revised using novel molecular and chemical forms of evidence. Marine biodiversity is extremely ancient and life probably arose in a marine environment between 4500 and 3500 Mya (million years ago), with chemoautotrophic prokaryotes and photosynthetic cyanobacteria appearing by 3000 Mya, and eukaryotic cells around 2000 Mya (Battistuzzi *et al.* 2004, Cavalier-Smith 2006). Thus, for most of the time life has existed on Earth, biodiversity was entirely marine and microbial. Microbes lack the capacity to exploit behavioural or morphological diversity, have limited capacity to engineer environments and drive co-evolutionary change, and their extremely large populations are effectively immune from extinction and other ecological processes driving evolutionary change and diversification. The consequence would have been extreme evolutionary stasis. The Proterozoic oceans were generally anoxic, sulphidic, turbid, and stratified, and largely physically driven. Life in the water column was dominated by cyanobacteria, the small size of which limited their rate of sinking and maintained a positive feedback whereby cyanobacterial dominance, turbidity, and anoxia were maintained (Butterfield 2010). Life on the seafloor was characterized by microbial mats (Bottjer 2005). Precisely when multicellular organisms evolved in the sea is unknown. Molecular evidence suggests that plants, animals and fungi diverged about 1500–1600 Mya, the basal animal phyla (Porifera, Cnidaria, Ctenophora) diverged about 1200–1500 Mya, nematodes diverged from the line leading to arthropods and chordates about 1200 Mya, and arthropods and deuterostomes around 1000 Mya (Wang *et al.* 1999; Pisani *et al.* 2004). These estimates indicate the presence of complex metazoan life long before the oldest currently known direct fossil evidence, which indicates the presence of sponges at least 635 Mya (Love *et al.* 2009). The earliest known fossils of complex multicellular organisms date from somewhere between 595 and 565 Mya (Narbonnel and Gehlingl 2003), forming the Ediacaran biota, a phylogenetically diverse assemblage of organisms, probably including animals, protists, and fungi with a simple ecology dominated by epibenthic osmotrophs, deposit feeders and grazers with few, if any, predators (Xiao and Laflamme 2008). The earliest evidence of motile animals is from 565 Mya (Liu *et al.* 2010). This biota flourished worldwide until disappearing at the onset of the Cambrian approximately 542 Mya.

Most major groups of complex animals first appeared as fossils sometime in the Cambrian, the so-called 'Cambrian explosion'. The apparent absence of fossils from before this time (in the absence of knowledge about the Ediacaran biota) and the apparently sudden appearance of so many types of life without forerunners greatly concerned Charles Darwin (1859), who saw these facts as the greatest objection to his theory of evolution by natural selection. The Ediacaran and early Cambrian are marked by some of the largest

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biogeochemical perturbations in Earth's history including major changes in carbon and sulphur cycling, iron geochemistry, phosphate deposition, and oceanic oxygenation (Butterfield 2010). Until recently, such observations have been used to explain the sudden appearance of diverse animal assemblages in the Cambrian in terms of changes in the Earth system providing conditions conducive to metazoan evolution, with a focus on atmospheric oxygenation (e.g. Knauth and Kennedy 2009; Payne *et al.* 2009). Recently this picture has changed, with the realization that animals have particular abilities and qualities that allowed them to fundamentally alter the Earth system, and to drive evolutionary change (Butterfield 2010; Meysman *et al.* 2006). The evolution of animal organization allowed organisms to achieve different sizes, large and small organisms could interact, so predation and competition became factors driving evolution which could then occur at a pace more rapid and at an order of magnitude greater than previously. Predation promoted the evolutionary advantage of defensive and strengthening structures such as biomineralized skeletons with bristles and spines, promoting co-evolution among predators and prey and, coincidentally, the probability of organisms becoming fossilized. Animal feeding and bioengineering activities facilitated and altered biogeochemical cycles. For example, animals feeding on plankton converted slow-sinking cells to rapidly sinking faecal pellets (Logan *et al.* 1995), providing a source of organic-rich sediment, and animals burrowing in sediment could exploit buried carbon as a food source, at the same time oxygenating sediments and altering their biogeochemistry (Meysman *et al.* 2006). Phytoplanktonic organisms increased in size and developed biomineralized structures, causing them to sink more rapidly and therefore to change export production from the surface ocean. Concentrating biomass into fewer but larger organisms increased the availability of free oxygen. The evolution of filter-feeding provided a mechanism for clearing microbial populations and providing a relatively non-turbid photic zone in which algal dominated primary production could take place. Thus the co-occurrence of a major shift in the Earth system and marine biodiversity, from a turbid, stratified, anoxic, microbially dominated Proterozoic ocean to a clear, oxygenated, algal, and animal-dominated Phanerozoic ocean, may mark the effects of life on Earth, rather than the other way around.

Many studies have used the documented occurrences of taxa in the fossil record to analyse global trends in marine diversity through deep time. A much repeated curve generated from these studies (see Sepkoski 2002) shows a rapid increase in marine diversity through the Cambrian and Ordovician (to 446 Mya), a plateau lasting until the end of the Permian (254 Mya), and then an apparently exponential increase to the present. However, concerns about the extent to which this pattern is artefactual have led to studies redrawing the Phanerozoic diversity curve using standardized methods (Alroy *et al.* 2008). These new studies tend to show that, although diversity varied through time, the overall trend was a slow increase, with levels in the Neogene (23–7.2 Mya) only slightly higher than the levels seen in the Devonian and Permian. Sepkoski (1981) identified three major 'Evolutionary Faunas'—the Cambrian, the Palaeozoic, and the Modern—which sequentially replaced each other. Despite these gradual changes, the repeating motif of life in clear shallow seas throughout the Phanerozoic and Cenozoic is of diverse and abundant reef-building organisms (archaeocyathid sponges, rudist bivalves, corals), epifauna and infauna (molluscs, brachiopods, bryozoans, crustaceans, echinoderms), and iconic top predators (anomolacaridids, eurypterids, nothosaurs, ichthyosaurs, plesiosaurs, diverse types of fish, especially sharks).

1.5 The distribution of marine biodiversity

Most of the Earth's surface (65–70%, depending on how one defines it) is covered by deep seawater (> 200 m). Life in the deep sea is diverse and specialized with organisms living in constant darkness at high pressures and low temperatures. However, the number and variety of organisms per unit area or volume is generally low compared to shallow coastal systems, with diversity tending to decline

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with increasing depth. It is undoubtedly true that the proportion of the actual deep-sea fauna which is known to science is lower than for other marine systems, but enough is known to estimate that the total number of species in the deep ocean is unlikely to exceed the number in shelf seas. It should also be borne in mind that life in the deep sea is comparably young, as for most of Earth's history the deep sea has been inhospitable to life.

Despite covering less than 10% of the Earth's surface, coastal seas and oceans deliver approximately 30% of marine production and 90% of marine fisheries (Millennium Ecosystem Assessment 2005). Some temperate ecosystems are, or at least were, among the most productive on Earth (Suchanek 1994). They are the burial site for 80% of organic matter, and 90% of sedimentary mineralization and nutrient cycling takes place there. They also provide the sink for up to 90% of the suspended load in the world's rivers, along with many associated contaminants (Gattuso *et al.* 1998). Marine biodiversity is highest in the coastal and shelf seas supported by the presence of diverse sedimentary habitats, rocky and biogenic reefs, kelp forests, and seagrass beds. Many of these habitats are highly specialized. For example, estuaries are generally inhabited by a limited number of specialist species that can tolerate reduced salinities. Generally those species that live in estuaries are marine in origin, and there is a gradient of declining biodiversity from full salinity water at the estuary mouth towards the freshwater at the head of the estuary. Another specialized set of organisms inhabits the intertidal zone which ranges from rocky to sediment shores, and exposure to waves is a crucial determinant of the community inhabiting a particular shore. Again, the vast majority of species are marine in origin and there tends to be a gradient of declining biodiversity from permanently submerged shallow waters at the base of the shore to higher parts of the shore which are only inundated rarely, or are only wetted by spray. The increasing gradients in marine biodiversity from the land to the open coastal and shelf seas contrasts strongly with our knowledge about the species present, which is greatest in estuaries and on coasts, and declines towards the open sea (Lotze and Worm 2009).

1.6 Human impacts on marine biodiversity

Modern humans evolved in Africa around 200 000 years ago with the earliest evidence of humans at the coast being from the Eritrean coast of Africa 125 000 years ago (Walter *et al.* 2000). Using handaxes and obsidian blades, these early people were harvesting and processing oysters and other marine resources (Bruggemann *et al.* 2004). A likely scenario for the geographical spread of early humans is that they followed the coasts, especially during periods of low sea levels when much of the continental shelves would have been exposed. By 70 000 years ago, humans had spread to north-west Africa, and by 50 000 years ago, humans had reached south-east Asia and Australia. Humans were present on the coasts of east Asia and Siberia around 30 000 years ago and reached southern Europe some 20 000 years ago. Humans then colonized North America about 15 000 years ago and South America 15–12 000 years ago. Thus, by around 10 000 years ago, humans were present on all the major continents (except Antarctica, unreachable and uninhabitable without appropriate technology), leaving only remote islands to be colonized. During roughly 200,000 years as hunter-gatherers the human population probably never exceeded 10 million (May 2010); however, the introduction of agriculture approximately 10 000 years ago allowed the population to increase rapidly before a number of factors such as infectious diseases slowed growth, and it was not until 1830 that numbers exceeded 1000 million. From there the population doubled to 1930 and passed 4000 million in 1970. Following the scientific and industrial revolution and the development of science-based medical understanding, the Earth's human population has reached 6700 million and continues to increase. To put these figures in context, for every human alive today, only 15 have lived before, and the human population has trebled in less than a lifetime (May 2010). Nearly 70% of all humans now live within 60 km of the sea, many of them in cities of > 10 million inhabitants, of which 75% are in the coastal zone. It is therefore unsurprising that the activities of human beings are likely to have a significant impact on coastal marine systems.

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Following the upsurge of interest in biodiversity in the early 1990s, many reviews were published outlining the state of marine biodiversity, in the open ocean (Angel 1993), sediments (Snelgrove 1999), the coastal zone (Ray 1991), coral reefs (Jackson 1991), rocky shores (Thompson *et al.* 2002), estuaries (Levin *et al.* 2001), temperate coastal seas (Suchanek 1994), and the deep sea (Grassle 1991), along with more general reviews of marine biodiversity (Gray 1997, Ray and Grassle 1991, National Research Council 1995). These reviews described biodiversity patterns in marine systems and discussed the threats faced by marine ecosystems and the organisms they support. The general conclusion was that the most severe threats to marine biodiversity are in the coastal zone and are a direct result of human population and demographic trends (Gray 1997), a conclusion supported and expanded in the comprehensive reviews of current state and trends in the Millennium Ecosystem Assessment (2005). Although many of those pressures acting on estuarine and intertidal biodiversity, particularly habitat loss and habitat modification or squeeze, can have extreme local effects on marine biodiversity, these are fairly predictable, and many of the species affected are tolerant and many may actually benefit, such as those which can use man-made structures as habitat. Restoration in such systems is also often possible.

At the other end of the spectrum, the effects of man's activities in deep sea systems may be widespread and some, such as the removal of predators or the accumulation of toxins or garbage, may alter food-webs. Localized effects, such as commercial fishing of slow-growing and long-lived fish in geographically limited areas such as seamounts and continental shelves, with concomitant damage to seabed organisms, may be severe. Commercial and technological developments are pushing the limits at which activities such as oil and gas extraction, or seabed mining for minerals, are viable ever deeper. Despite all of this, the direct effects of man's activities on deep-sea biodiversity as a whole are unlikely to be as severe as in shallower waters.

It is in the intermediate shallow shelf and coastal waters, where marine biodiversity evolved and has flourished throughout most of the history of life, that the effects of pressures from human activities are highest. It is no coincidence that recorded extinctions such as the Great Auk, Stellar's Seacow, the Caribbean Monk Seal, and ecological extinctions such as many large species of skates and rays and commercially fished populations of other fish, or major changes in ecosystems such as the demise of coral reefs and cod, are to be found in such waters. Species have not evolved mechanisms to cope with sudden change and the effects of pressures are relatively poorly understood. It used to be thought that the fluid nature of life in the sea, with widespread dispersal, could insure most marine organisms against total extinction. More recently it has become clear that there are many invisible barriers to dispersal in the sea, so species may be more vulnerable than was thought. Sadly we have no idea of the number of populations or species that may already have been lost or how many will be lost in the future.

The major threats to marine biodiversity are numerous: habitat degradation, fragmentation and loss, climate change and other large-scale atmospheric changes such as ozone depletion, direct and indirect effects of fishing and other forms of overexploitation, nutrient runoff, mineral extraction, pollution and marine litter, species relocations and invasions, watershed alteration and the physical alteration of coasts, tourism, limited knowledge and willingness to conserve marine biodiversity, and legal complexities in dealing with marine issues. Realizing that, even among professionals, knowledge about what marine biodiversity ought to be like in the absence of human intervention was generally missing (Pauly 1995, Jackson 2001), and in recognition that a research strategy based on short-term experiments or observations is unlikely to be helpful in addressing this issue, a number of studies have subsequently taken a multidisciplinary approach, incorporating elements of palaeontology, archaeology, history, and ecology, to describe long-term trends in marine biodiversity. Using a range of different data sources, a strategy based on the facts that different parts of the world were colonized by people at different times, and that in different places cultural development followed similar paths, a clear pattern of degradation has been shown in

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a diverse range of marine communities and ecosystems: coastal (Jackson *et al.* 2001), coral reefs (Pandolfi *et al.* 2003), estuaries and coastal seas (Lotze *et al.* 2006), large marine animals (Lotze and Worm 2008). These studies, summarized and expanded by Jackson (2008), point to the fact that even small human populations may have severe effects on marine biodiversity, and that although humans have been degrading marine systems for a very long time, the recent increase in human population and its consequences have led to a step-shift in the rate and extent of degradation.

A primary cause of biodiversity degradation is the removal of large animals and the consequent disruption of food-webs through trophic cascades. In the industrial age (circa 1750 to present day), these effects may be very rapid and widespread (Myers and Worm 2003). Large-scale exploitation of marine resources probably did not occur in Europe before AD 1000, and when it did it may have been a response to the rise of urbanization and the overexploitation and degradation of freshwater ecosystems (Barrett *et al.* 2004). Major exploitation of marine resources elsewhere in the world largely followed the arrival of Europeans. The consequent rapid decline of marine ecosystems is therefore remarkable with the majority of coastal ecosystems now being severely degraded compared to long-term historical baselines. Through human activities, complex ecosystems are being transformed into monotonous level bottoms, clear and productive seas into anoxic dead zones (Vaquer-Sunyer and Duarte 2008), complex food-webs with large apex predators into microbially dominated systems. These changes, along with similar but better-known changes in terrestrial and freshwater ecosystems, have led to some proposing a new term, the 'Anthropocene epoch', to separate the world as it is now from the world as it was until recently. As Jackson (2008) points out, we do not know what organisms will benefit from and come to dominate marine ecosystems in the near future, but there is already evidence that jellyfish may be among them (Richardson *et al.* 2009) along with microbes, including some that may have the potential to be toxic or cause disease (Jackson 2008). We are only beginning to realize the extent of the damage we have already done, are doing, and are likely to do, and what the likely consequences will be (Worm *et al.* 2006). For the anthropogenic impacts that are performed at local scales—e.g. fishing, pollution, habitat destruction—these trends may still be reversible. However, the impacts associated with climate change are likely to be far more ubiquitous and long lasting.

1.7 The relationship between global climate and marine biodiversity

Close examination of the geological record provides chilling evidence that Planet Earth has historically been vulnerable to climate-induced biodiversity loss on a huge scale. Prior to large extinction events in the late Permian/early Triassic, early Jurassic, middle Cretaceous, and late Paleocene, major volcanic events occurred and the carbon-isotope record confirms that during these periods atmospheric CO_2 concentrations rose considerably and stayed high for hundreds of thousands of years. Although the current rise in atmospheric CO_2 levels is not volcanically driven, the consequences could essentially be similar. The largest of all climatically driven extinction events occurred 251 Mya at the end of Permian period. During this event, 80% of all marine species and 49% of all marine families went extinct (Raup and Sepkoski 1982; Stanley and Yang 1994). It is suggested that the environmental changes associated with the late-Permian extinction event were triggered by huge quantities of volcanically produced CO_2 and methane that drove rapid global warming. As the oceans warmed, they carried less oxygen and these conditions encouraged the growth of bottom-dwelling anaerobic bacteria. These bacteria produced large quantities of hydrogen sulphide (H_2S), and eventually the chemocline (the division between oxygen-rich surface water and H_2S-rich deep water) rose up into the photic zone allowing the proliferation of green and purple photosynthesizing sulphur bacteria while oxygen-breathing organisms suffocated. The H_2S also diffused into the air where it killed animals and plants on land. Once into the upper atmosphere, the H_2S destroyed the ozone layer and the Sun's ultraviolet radiation caused even more extinctions (Kump *et al.* 2005).

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Whilst it is difficult to imagine that such a fate could befall modern Earth, it is worth reflecting on the atmospheric levels of CO_2 that correspond to some of these large extinction events. At the start of the late-Permian event, the concentration was around 3000 parts per millon (ppm), almost an order of magnitude higher than today's value of 385 ppm. However, the smaller yet still significant extinctions which occurred at the end of the Paleocene and at the end of the Triassic began when atmospheric CO_2 levels were much lower, around 1000 ppm. If current increases in atmospheric CO_2 levels continue as projected, we could be looking at levels approaching 1000 ppm by the end of the next century (IPCC 2007).

1.8 Could marine biodiversity be facing large-scale climate-induced extinction?

At the start of the industrial revolution (circa 1750), the atmospheric concentration of carbon dioxide (CO_2) in the atmosphere was around 280 ppm. In the subsequent 250 years, this has risen rapidly to a current value of more than 385 ppm (Feely *et al.* 2004). This increase has been driven primarily by the burning of fossil fuels, wide scale deforestation, changes in land use, and the manufacture of cement (IPCC, 2007). Once in the atmosphere CO_2, along with other 'greenhouse' gases such as methane and nitrous oxide, retains the Sun's heat and warms up the Earth's lower atmosphere. As 65–70% of the planet's surface is covered by water, much of this heat is being absorbed by the oceans, causing mean global sea-surface temperatures to have increased at a rate of 0.13 °C per decade over the past 30 years (IPCC). In addition to causing this warming, the process of radiative forcing (the entrapment of the Sun's energy in the lower atmosphere) is supplying the energy which is driving a number of other climate-change phenomena. These include increased average wind velocities and storm frequency, increased global average precipitation and riverine runoff, and enhanced sea-level rise through thermal expansion and melting of ice. All of these physical effects could have a significant indirect impact on the marine environment by altering ocean currents, changing the supply and cycling of key nutrients, and by reducing seawater salinity. For a summary of these impacts on a range of marine ecosystems, see Brierley and Kingsford (2009). It must also be considered that climate change impacts will be geographically variable with some locations, such as the Arctic Ocean, experiencing much larger changes than the global average. An appreciation of the potential severity of these impacts has lead some authors to predict that climate change will be the main driver of global biodiversity loss over the next century (Thomas *et al.* 2005).

As organisms look to optimize the structural and kinetic coordination of molecular, cellular and systemic process, they are constrained to exist within a physiologically prescribed thermal window (Figure 1.1). At the edges of these thermal windows an organism will experience decreased performance with respect to many functions: growth, reproduction, immune response (Pörtner and Farrell 2008). At these edges, biological interactions, such as predation and competition, will determine which species are able to persist and therefore the structure and diversity of the resident community. Consequently, changes in seawater temperature have the potential to alter the performance of resident species, change the competitive balance between organisms, and increase the likelihood of invasive species entering an area. All of this will modify the strength and direction of the key biological interactions which underpin biodiversity. However, temperature is not the only environmental stressor involved in this process. Figure 1.1 also indicates that additional environmental stressors, such as hypercapnia, hypoxia, and salinity change, can narrow an organism's thermal window and therefore increase its susceptibility to extreme temperature events.

Climate change could also promote biodiversity loss through indirect impacts. In many marine systems the activities of individual species can be instrumental in maintaining high levels of biodiversity in the associated fauna (see review by Widdicombe and Austen 2005). This biological control can be exerted through reducing competitive exclusion *via* density dependent predation or consumption, i.e. 'keystone species' (Paine 1966) or through habitat modification and creation of

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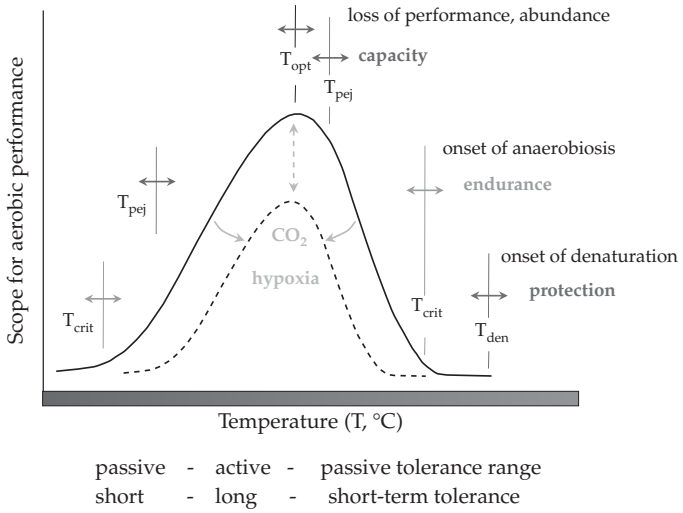

Figure 1.1 Relationship between organism performance and environmental temperature. From Pörtner and Farrell (2008), used with permission from AAAS.

environmental heterogeneity via processes such as reef building and bioturbation, i.e. 'ecosystem engineering' (Jones *et al.* 1994; Lawton 1994). Biogenic habitats such as coral reefs, mearl beds, and mussel banks are considered to be particularly important as 'hot spots' for biodiversity. However, many of these important habitats rely heavily on the health and survival of thermally sensitive, calcifying species which are considered to be at high risk from climate change (Raven *et al.* 2005). In addition, many keystone species and ecosystem engineers are predominantly large organisms and it might be considered that with larger organisms having generally lower metabolic rates than smaller organisms, increasing body size should enhance tolerance to increased temperature, hypercapnia, and anoxia. However, the current change in environmental conditions associated with climate change is more of a chronic pressure than an acute one with changes occurring over a timescale of tens to hundreds of years. Therefore, the long-term survival of any species is more likely to be determined by its ability to adapt rather than the ability of individuals to persist. Consequently, large animals with longer generation times will be at an adaptive disadvantage. There is already some evidence from the geological record that marine organisms do become smaller in conjunction with many of the previous mass extinction events (Twitchett 2007); a phenomenon known as the 'Lilliput effect'. It should also be considered that a reduction in the top–down biological control of biodiversity could come about not just through the complete loss of important species, but also through changes in the rates of ecologically important activities, such as growth, feeding, bioturbation, and burrow irrigation.

1.9 Additional impacts of CO_2 on the marine environment

Whilst the measured rise in atmospheric CO_2 levels and global temperatures are dramatic, these increases would have been even greater were it not for the fact that almost half of all the anthropogenic CO_2 emitted over the last 250 years has been taken up by the world's oceans (Sabine *et al.* 2004). It is natural for atmospheric CO_2 levels to fluctuate through time creating CO_2 gradients between the atmosphere and the ocean as the air and seawater gradually equilibrate. These gradients will subsequently drive the oceans to absorb CO_2 during times of rising atmospheric concentrations and to release it during times of falling atmospheric concentrations. As we are currently in a period of rising

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atmospheric concentrations, CO_2 diffuses passively into ocean surface waters where it reacts with water molecules to form carbonic acid (H_2CO_3), a weak acid that rapidly dissociates to produce hydrogen (H^+) and bicarbonate (HCO_3^-) ions. This increase in H^+ ion concentration would cause seawater pH to fall—i.e. the acidity to increase—was it not for the saturation of surface waters with carbonate cations (CO_3^{2-}). These cations react with the H^+ ions to form more HCO_3^- and this entire reaction ($CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow HCO_3^- + H^+ \leftrightarrow CO_3^{2-} + 2H^+$) is referred to as the carbonate buffering system; a process by which surface ocean pH has remained relatively stable (between 8.0 and 8.3 pH units) for the last 25 million years.

The current concentration of atmospheric CO_2 is not particularly unusual and is, in fact, still quite low compared to some periods in Earth's recent history, so it was assumed that the carbonate buffering system would continue to maintain seawater pH at 'normal' levels despite rising emissions. However, this assumption was challenged in 2003 when model predictions were used to show that, compared to pre-industrial times, ocean surface water pH had fallen by 0.1 pH unit, indicating a 30% increase in the concentration of H^+ ions (Caldeira and Wickett 2003); a phenomenon now widely recognized and referred to as 'the other CO_2 problem' or 'ocean acidification'. It is now clear that the major cause of ocean acidification is not the total amount of CO_2 entering the oceans, but the unprecedentedly short time over which this uptake is occurring. Atmospheric CO_2 concentrations are now increasing by more than 2 ppm per year. This rate exceeds that predicted by the Intergovernmental Panel on Climate Change (IPCC) under a 'worst case scenario' (IPCC 2007) and is nearly 100 times greater than any historical, naturally induced increase (Blackford and Gilbert 2007). As the carbonate buffering system relies on the availability of CO_3^{2-} cations, which are primarily derived from relatively slow geological process such as the weathering of carbonate rich rocks or the dissolution of carbonate sands, the Earth's oceans are rapidly becoming more acidic and experiencing increasing hypercapnia (elevated levels of dissolved CO_2). Recently published observational data have since confirmed these model conclusions with declining values for pH being reported in a number of long running oceanographic time series, e.g. the Hawaiian Ocean Time-series (HOT) in the sub-tropical North Pacific (Brix *et al.* 2004), the European Station for Timeseries in the Ocean, Canary Islands (ESTOC) in the sub-tropical (East) North Atlantic (Santana-Casiano *et al.* 2007), the Bermuda Atlantic Time-Series (BATS) in the sub-tropical (West) North Atlantic (Bates and Peters 2007), and the Iceland Sea Polar (Nordic Seas) in the North Atlantic (Olafsson *et al.* 2009). All of these have reported a consistent trend for declining pH of between 0.0012 and 0.0024 pH units per year. If these trends continue, the pH of surface waters could be predicted to fall by on average up to 0.4 units before 2100 and by 0.7 units before 2250.

The potential physiological impacts of high levels of CO_2 on the health and function of marine species and communities have been detailed in a number of recent reviews (Seibel and Walsh 2001, 2003; Pörtner *et al.* 2004, 2005; Fabry *et al*. 2008; Widdicombe and Spicer 2008). In summary, when marine organisms are exposed to low pH seawater, the primary physiological effect is a decrease in the pH or an 'acidosis' of the extracellular body fluids such as blood, haemolymph, or coelomic fluid. In some species, this extracellular acidosis is fully compensated for as levels of extracellular bicarbonate are increased by either active ion transport processes in the gills or through passive dissolution of a calcium carbonate shell or carapace (see Widdicombe and Spicer 2008, and references therein). However, in other species from a variety of different taxa, such as the mussel *Mytilus edulis* (Michaelidis *et al.* 2005), the crabs *Callinectes sapidus* (Wood and Cameron 1985) and *Chionoecetes* tanneri (Pane and Barry 2007), and the sea urchin *Psammechinus* miliaris (Miles *et al.* 2007), studies have reported only partial, or no, compensation in the extracellular acid-base balance. Clearly if some species are physiologically better equipped to cope with elevated levels of CO_2 than others, the potential for species extinctions and biodiversity loss exists. At the phylum level, early evidence would suggest that echinoderms will be more vulnerable than molluscs, then crustaceans, with annelids showing the greatest tolerance to hyper-

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capnia and acidification (Widdicombe *et al.* 2009). So in a future, more acidic ocean, marine communities could be made up of species from a limited number of tolerant taxonomic groups. Whilst this may or may not reduce the total number of species in any given area, as tolerant species could potentially move in to exploit vacated niches, it would certainly reduce taxonomic richness as well as species diversity both regionally and globally. However, even within taxonomic groups, variability in tolerance can exist between even closely related species with this variability seemingly linked to key elements of an organism's lifestyle. Organisms that already exist in habitats regularly exposed to highly variable levels of temperature, and CO_2 may be more likely to possess the physiological mechanisms necessary to cope with rapid changes in environmental conditions than organisms from areas with more stable conditions, such as the polar oceans, deep sea, or well oxygenated sands. For example, Pane and Barry (2007) reported marked differences in the acid-base responses to acute seawater hypercapnia between the deep sea Tanner crab *Chionoecetes tanneri* and the shallow water Dungeness crab *Cancer magister*. Also, as ocean warming and acidification will be most severe in shallow surface waters, pelagic organisms or benthic species that rely on a pelagic reproductive stage could be more vulnerable than species with direct benthic development. It seems clear that the likelihood that a species will be lost from an area as a result of increased environmental stress could be determined by both its phylogeny and its ecology. Consequently, the potential exists for climate change to reduce the total number of marine species, the taxonomic richness of the communities which remain and the diversity of functions that these communities can perform.

1.10 Hypoxia and 'dead zones'

Oxygen is a prerequisite for sustaining multicellular life in the oceans. However, dissolved oxygen concentrations are declining in a growing number of coastal areas. Over recent decades the number of coastal sites where hypoxia (DO concentration ≤ 2 mg O_2/litre) has been reported has increased exponentially with Vaquer-Sunyer and Duarte (2008) estimating this increase to be around 5.5% every year. This hypoxia can not only cause large-scale mortality in metazoan animals, it can also exert sub-lethal stresses which inhibit growth and reproduction. Hypoxic stress can also force organisms from their normal habitats leaving them open to increased pressure from predation or competition. All of these processes can cause a loss of biodiversity and create the areas which are now termed 'dead zones' (Rabalais *et al.* 2001). The primary cause of this proliferation in dead zones is increasing eutrophication (Diaz and Rosenberg 2008), fuelled by an enormous growth in coastal cities and the increasing use of industrially produced nitrogen fertilizer by modern agriculture. As is likely to be the case for acidification, marine organisms exhibit a range of vulnerabilities to hypoxia and so the impacts on biodiversity are likely to be similar with a loss of species, taxonomic, and functional diversity.

There is strong evidence to suggest that climate change could exacerbate the problem of hypoxia in coastal marine ecosystems. Firstly, increased precipitation could increase riverine runoff and therefore transport more nutrients to the coastal zone. For example, Justić *et al*. (1996) suggested that under future climate scenarios, there would be a 20% increase in river discharge from the Mississippi and the nutrients associated with this would fuel a 50% increase in primary production and a decrease in subpycnocline dissolved oxygen of up to 60%. Secondly, over the range 0 °C to 15 °C the relationship between temperature and dissolved oxygen concentrations in seawater is approximately linear, with DO decreasing by about 6% for every 1° C rise in temperature. Increasing seawater temperatures could also promote thermal stratification and promote phytoplankton production. Finally, changes in wind strength and direction, or the frequency and intensity of storms could have contrasting effects. Off the western continental margins of South America, Africa, and India, offshore winds drive the upwelling of nutrient-rich bottom water and promote productivity. This process already creates large oxygen-minimum zones in these areas and any increase in wind strength could expand the

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extent of these zones. In other areas however, strong winds and intense storms could act to reduce hypoxia by removing stratification and enhance mixing of oxygen into deeper waters.

1.11 Summary

Biodiversity loss is happening and has been happening for hundreds of years. It is being driven by a multitude of anthropogenic stresses at a variety of spatial (local, regional, and global) and temporal (days to decades to centuries) scales. These stresses will act simultaneously to reduce taxonomic and functional diversity. The challenge for society now is to manage our exploitation of, and interaction with, the marine environment in a sustainable way so as to halt biodiversity loss and, where possible, promote recovery. However, the tools and policies necessary for such management must be based on a solid foundation of biodiversity-related science. As the following chapters will illustrate, there is substantial evidence that biodiversity is important for the delivery of ecosystem function; however, the mechanisms by which this occurs are still uncertain. Only by investing sufficient time and resources in studies which explore the fundamental mechanisms which underpin the biodiversity ecosystem function relationship will we generate the understanding necessary to predict the long-term impacts of man's activities on the marine environment.

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